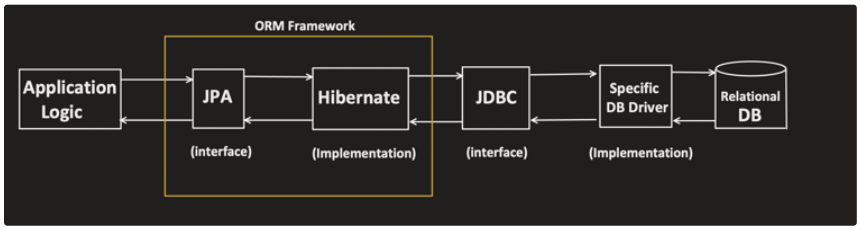
JPA-PART1



Before going to JPA, lets recall JDBC

JDBC (Java Database Connectivity) provides an Interface to:

- Make connection with DB
- Query DB
- and process the result

Actual implementation is provided by Specific DB Drivers.

For example:

Concept && Coding

MySQL

- Driver : Connector/J
- Class : *com.mysql.cj.jdbc.Driver*

PostgreSQL

- Driver : PostgreSQL JDBC Driver
- Class : *org.postgresql.Driver*

H2 (in-memory)

- Driver : H2 Database Engine
- Class : *org.h2.Driver*

Using **JDBC** without Springboot

```
public class DatabaseConnection {

    public Connection getConnection() {
        try {

            // H2 Driver loading
            Class.forName("org.h2.Driver");

            // Establish connection with DB
            return DriverManager.getConnection(
```

```
                    "jdbc:h2:mem:userDB",
                    "sa",
                    ""
            );
        }
        catch (ClassNotFoundException | SQLException e) {
            //handle exception
        }

        return null;
    }
}

```

```
public class UserDAO {

    public void createUserTable() {
        try {
            Connection connection = new DatabaseConnection().getConnection();
            Statement statementQuery = connection.createStatement();
            String sql = "CREATE TABLE users(user_id INT AUTO_INCREMENT PRIMARY KEY, user_name VARCHAR(100), age INT)";
            statementQuery.executeUpdate(sql);
        }
        catch (SQLException e) { /** handle exception **/ }
        finally { /** close statementQuery and db connection **/ }
    }

    public void createUser(String userName, int userAge) {
        try {
            Connection connection = new DatabaseConnection().getConnection();
            String sqlQuery = "INSERT INTO users(user_name, age) VALUES (?, ?)";
            PreparedStatement preparedQuery = connection.prepareStatement(sqlQuery);
            preparedQuery.setString(1, userName);
            preparedQuery.setInt(2, userAge);
            preparedQuery.executeUpdate();
        }
        catch (SQLException e) { /** handle exception **/ }
        finally { /** close preparedQuery and db connection **/ }
    }

    public void readUsers() {
        try {
            Connection connection = new DatabaseConnection().getConnection();
            String sqlQuery = "SELECT * FROM users";
            PreparedStatement preparedQuery = connection.prepareStatement(sqlQuery);
            ResultSet output = preparedQuery.executeQuery();
            while (output.next()) {
                String userDetails =
                        output.getInt("user_id") + ":" +
                        output.getString("user_name") + ":" +
                        output.getInt("age");

                System.out.println(userDetails);
            }
        }
        catch (SQLException e) { /** handle exception **/ }
        finally { /** close preparedQuery and db connection **/ }
    }
}

```

Concept && Coding

If we see above example:

- Connection
- Statement
- PreparedStatement
- ResultSet etc.

All are interfaces which JDBC provide and each specific driver provide the implementation for it.

But there are so much of **BOILERCODE** present like:

- Driver class loading
- DB Connection Making
- Exception Handling
- Closing of the DB connection and other objects like Statement etc.
- Manual handling of DB Connection Pool

Etc..

Using JDBC with Springboot

pom.xml

Concept && Coding

```
<dependency>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-jdbc</artifactId>
</dependency>

<dependency>
    <groupId>com.h2database</groupId>
    <artifactId>h2</artifactId>
    <scope>runtime</scope>
</dependency>

```

Springboot provides *JdbcTemplate* class, which helps to remove all the boiler code.

```
@Component
public class UserService {

    @Autowired
    UserRepository userRepository;

    public void createTable() {
        userRepository.createTable();
    }

    public void insertUser(String userName, int age) {
        userRepository.insertUser(userName, age);
    }

    public List<User> getUsers() {
        List<User> users = userRepository.getUsers();
        for(User user : users) {
            System.out.println(user.userId + ":" + user.getUserName()
+ ":" + user.getAge());
```

```
        }
        return users;
    }
}

```

```
@Repository
public class UserRepository {

    @Autowired
    JdbcTemplate jdbcTemplate;

    public void createTable() {
        jdbcTemplate.execute(
                "CREATE TABLE users (user_id INT AUTO_INCREMENT PRIMARY KEY, " +
                "user_name VARCHAR(100), age INT)"
        );
    }

    public void insertUser(String name, int age) {
        String insertQuery = "INSERT INTO users (user_name, age) VALUES (?, ?)";
        jdbcTemplate.update(insertQuery, name, age);
    }

    public List<User> getUsers() {
        String selectQuery = "SELECT * FROM users";
        return jdbcTemplate.query(selectQuery, (rs, rowNum) -> {
            User user = new User();
            user.setUserId(rs.getInt("user_id"));
            user.setUserName(rs.getString("user_name"));
            user.setAge(rs.getInt("age"));
            return user;
        });
    }
}

```

Concept && Coding

application.properties

```
spring.datasource.url=jdbc:h2:mem:userDB
spring.datasource.driver-class-name=org.h2.Driver
spring.datasource.username=sa
spring.datasource.password=
spring.h2.console.enabled=true

```

```
public class User {

    int userId;
    String userName;
    int age;

    //getters and setters
}

```

Driver class loading -

JdbcTemplate load it at the time of application startup in DriverManager class.

DB Connection Making -

jdbcTemplate takes care of it, whenever we execute any query.

Exception Handling -
in Plain JDBC, we get very abstracted '*SQLException*' but in jdbcTemplate, we get granular error like DuplicateKeyException, QueryTimeoutException etc.. (defined in org.springframework.dao package).

Closing of the DB connection and other resources -

when we invoke update or query method, after success or failure of the operation, jdbcTemplate takes care of either closing or return the connection to Pool itself.

- **Manual handling of DB Connection Pool** -
  Springboot provides default jdbc connection pool i.e. *'HikariCP'* with Min and Max pool size of 10 And we can change the configuration in *'application.properties'*

spring.datasource.hikari.maximum-pool-size=10

spring.datasource.hikari.minimum-idle=5

We can also configure different jdbc connection pool if we want like below:

```
@Configuration
public class AppConfig {

    @Bean
    public DataSource dataSource() {
        HikariDataSource dataSource = new HikariDataSource();
        dataSource.setDriverClassName("org.h2.Driver");
        dataSource.setJdbcUrl("jdbc:h2:mem:userDB");
        dataSource.setUsername("sa");
        dataSource.setPassword("");
        return dataSource;
    }
}

```

Concept && Coding

JdbcTemplate frequently used methods

Method Name	**Use For**	**Sample**
update(String sql, Object... args)	Insert Update Delete	String insertQuery = "INSERT INTO users (user_name, age) VALUES (?, ?)"; int rowsAffected = jdbcTemplate.update(insertQuery , "X", 27);

		String updateQuery= "UPDATE users SET age = ? WHERE user_id = ?";
		int rowsAffected = jdbcTemplate.update(updateQuery, 29, 1);
		
		String deleteQuery= "DELETE FROM users WHERE user_id = ?";
		int rowsAffected = jdbcTemplate.update(deleteQuery, 1);

update(String sql, PreparedStatementSetter pss)	Insert
	Update
	Delete
		String insertQuery= "INSERT INTO users (user_name, age) VALUES (?, ?)";
		
		jdbcTemplate.update(insertQuery, (PreparedStatement ps) -> {
		
		ps.setString(1, "X");
		
		ps.setInt(2, 25);
		
		});

Concept && Coding

		String updateQuery= "UPDATE users SET age = ? WHERE user_id = ?";
		
		jdbcTemplate.update(updateQuery, (PreparedStatement ps) → {
		
		ps.setString(1, 29);
		
		ps.setInt(2, 1);
		
		});
		
		String deleteQuery= "DELETE FROM users WHERE user_id = ?";
		
		jdbcTemplate.update(deleteQuery, (PreparedStatement ps) → {
		
		ps.setInt(1, 2);
		
		});

query(String sql, RowMapper<T> rowMapper)	Get multiple Rows	List<User> users = jdbcTemplate.query("SELECT * FROM users", (rs, rowNum) → {

Concept && Coding

		User user = new User();
		
		user.setUserId(rs.getInt("user_id"));
		
		user.setUserName(rs.getString("user_name"));
		
		user.setAge(rs.getInt("age"));
		
		return user;
		
		});
		
		Concept && Coding

queryForList(String sql, Class<T> elementType)	Get Single Column of Multiple Rows	List<String> userNames = jdbcTemplate.queryForList("SELECT user_name FROM users", String.class);
queryForObject(String sql, Object[] args, Class<T> requiredType)	Get single Row	User user = jdbcTemplate.queryForObject("SELECT * FROM users WHERE user_id = ?", new

		Object[]{1}, User.class);
queryForObject(String sql, Class<T> requiredType)	Get Single Value	int userCount = jdbcTemplate.queryForObject("SELECT COUNT(*) FROM users", Integer.class);

Concept && Coding